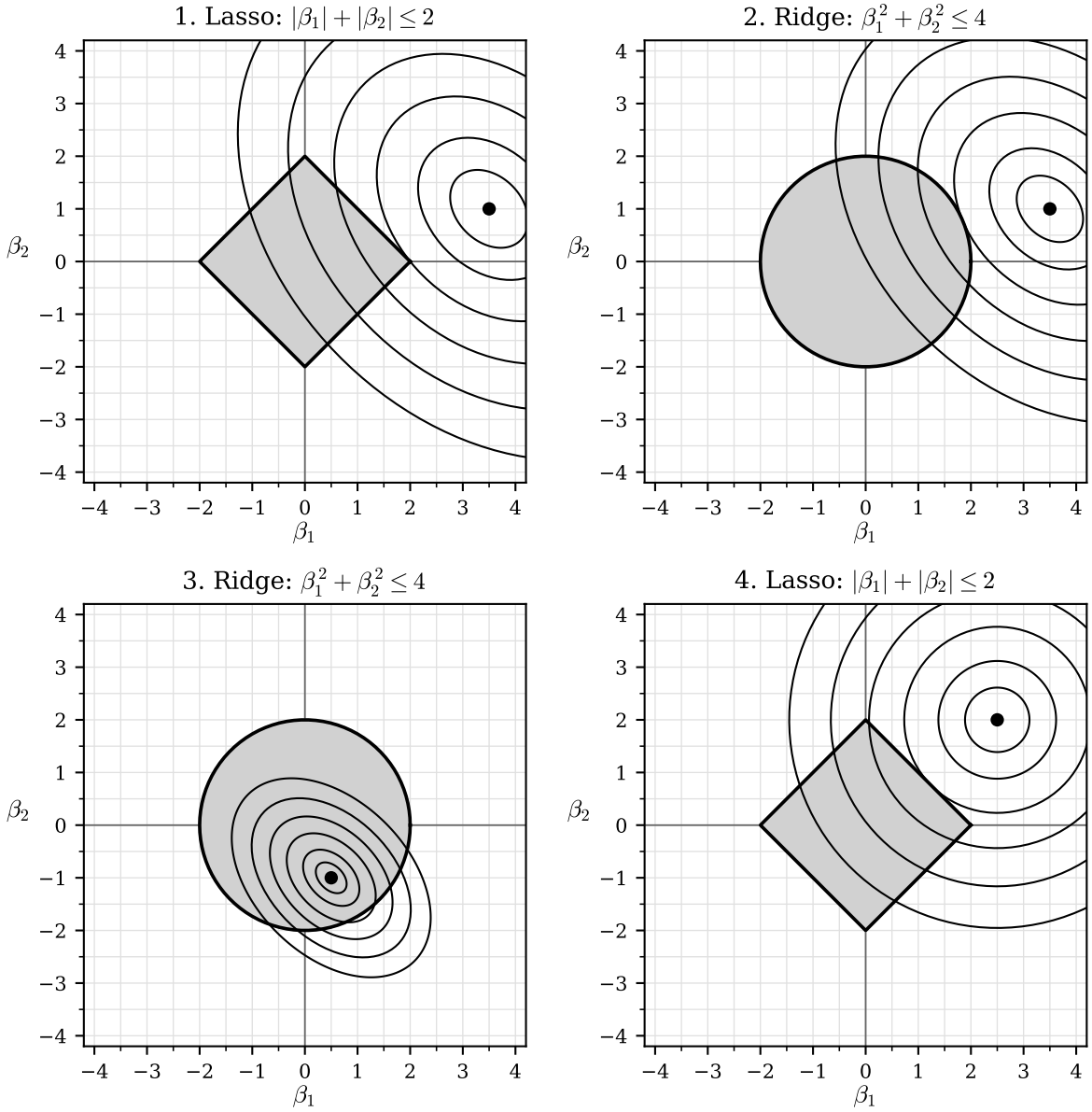
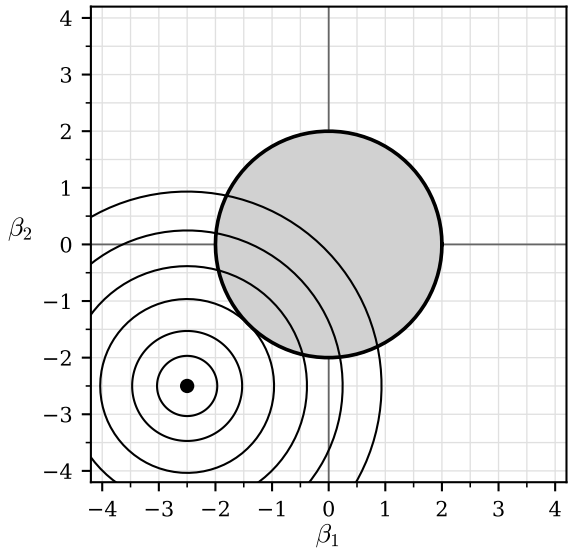
Intro to Machine Learning
Quiz 2 Practice: Constraint Pictures

Each panel shows contours of the RSS for a model with two predictors (ellipses around the OLS estimate, the solid dot) and a shaded hard-constraint region: the lasso, $|\beta_1| + |\beta_2| \leq 2$, or ridge, $\beta_1^2 + \beta_2^2 \leq 4$. The constrained estimate $(\hat{\beta}_1, \hat{\beta}_2)$ is the point of the shaded region with the smallest RSS.
For each panel: (a) mark the constrained estimate with an X; (b) read off its approximate coordinates (the grid lines are every 0.5); (c) say whether either coefficient is exactly 0.
Hint: RSS is constant along each ellipse and grows as the ellipses get bigger. Not every contour is drawn, so imagine the ones in between.

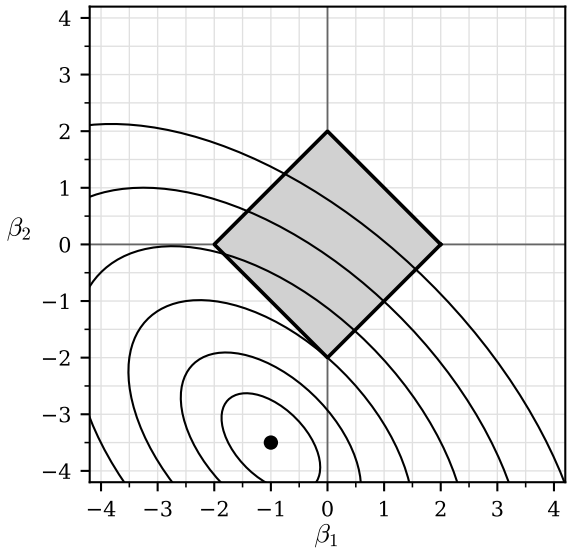


1. $(\hat{\beta}_1, \hat{\beta}_2) \approx$ _____ Is either coefficient exactly 0? _____
2. $(\hat{\beta}_1, \hat{\beta}_2) \approx$ _____ Is either coefficient exactly 0? _____
3. $(\hat{\beta}_1, \hat{\beta}_2) \approx$ _____ Is either coefficient exactly 0? _____
4. $(\hat{\beta}_1, \hat{\beta}_2) \approx$ _____ Is either coefficient exactly 0? _____

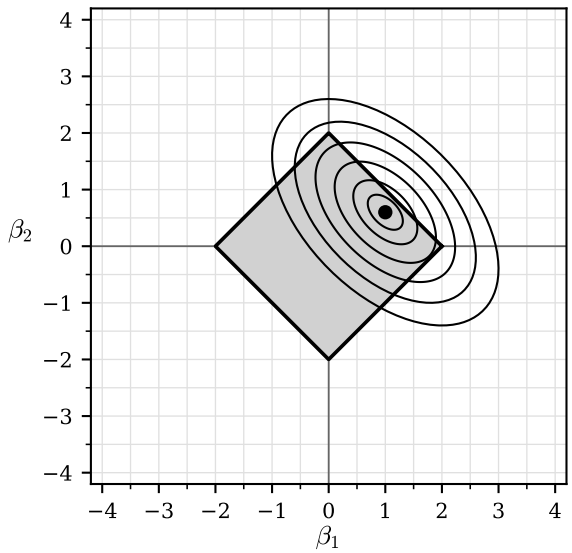
5. Ridge: $\beta_1^2 + \beta_2^2 \leq 4$



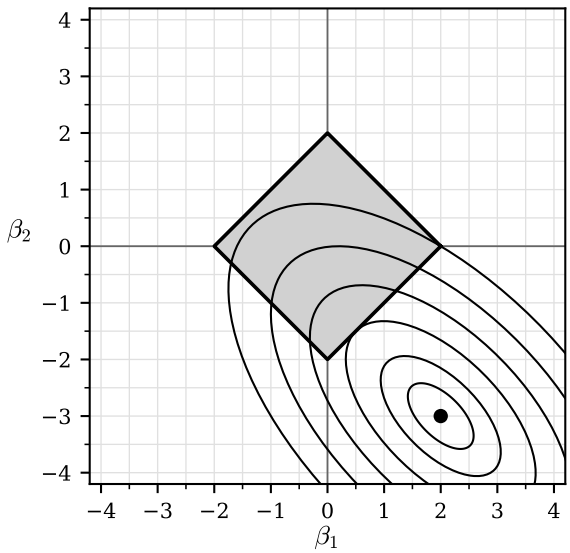
6. Lasso: $|\beta_1| + |\beta_2| \leq 2$



7. Lasso: $|\beta_1| + |\beta_2| \leq 2$



8. Lasso: $|\beta_1| + |\beta_2| \leq 2$



5. $(\hat{\beta}_1, \hat{\beta}_2) \approx$ _____ Is either coefficient exactly 0? _____
6. $(\hat{\beta}_1, \hat{\beta}_2) \approx$ _____ Is either coefficient exactly 0? _____
7. $(\hat{\beta}_1, \hat{\beta}_2) \approx$ _____ Is either coefficient exactly 0? _____
8. $(\hat{\beta}_1, \hat{\beta}_2) \approx$ _____ Is either coefficient exactly 0? _____